# Rupali Bhatnagar

Senior Software Engineer

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## Work Experience

| ServiceNow | Sr. Software Engineer (Security Operations) | Mar'22 – May'25 |
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Individual contributor on Security Operations and Vulnerability Response, building on the ServiceNow© platform. Underlined features link to ServiceNow's public documentation.

- **Exception Management — Approval Rule Engine** for vulnerability exceptions — designed a multi-level, dynamic approval rule engine on ServiceNow Flow Designer supporting deferment and exception requests across differing enterprise policies; Flow Designer approvals shipped in **v15.0.2**. **Reduced approval time by 40%.**
- **Penetration Testing Request Catalog & Workflow Automation** — built the request catalog and automated intake, triage and routing for penetration-test requests across **50+ subsystems**, **cutting manual coordination by more than 30%** — the testing itself stays expert and manual by design.
- **Background Job Processing Framework** — architected the multithreaded framework for long-running tasks that previously strained system resources, partitioning work into parallel chunks, with migration via a configuration interface. Rollup jobs were converted to multithreaded background jobs in **v18.2.5**:
  - Multithreaded execution — **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
  - Job status tracking and cancellation — **100% visibility** on executing and errored tasks.
  - Configurable **retry logic and dead-letter queuing** for failed jobs.
- **Real-Time Data Aggregation Engine** — high-performance aggregation framework for near real-time multi-dimensional analytics, powering key internal dashboards over **100K+ records in an 8-hour window**.

| ServiceNow | Software Engineer (Security Operations) | Apr'19 – Mar'22 |
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- **Assignment Rule Engine for Vulnerability Remediation** — cache-optimised engine prioritising infrastructure vulnerabilities and policy violations and routing **vulnerable items (VITs)** to the correct remediation groups via the Glide platform. Used daily by ITIL users; handles **1M+ vulnerable items daily**.
- **Assignment Recommendations on Predictive Intelligence** — classification models on the **Predictive Intelligence** framework with feature engineering and a reviewer-facing recommendation interface across **vulnerable items and vulnerability groups**; shipped in **v12.1.4**. **20% increase in task allocation accuracy.**
- **CISO & Vulnerability Dashboards** — led design and development of the CISO dashboard and supporting vulnerability dashboards, defining the key metrics and structuring the data pipeline. Used on a regular basis by **50+ organisations**.
- **Vulnerability Source Integration** — engineered integrations with **Veracode, Qualys, Tenable and Microsoft TVM**, building the **ETL pipelines** that standardise **500K+ vulnerability records/day** into a common model, with error handling and retry for reliable ingestion into downstream remediation workflows.
- **POA&M Generator** — auto-generated **Plans of Action & Milestones**, the remediation-plan artifact required under **NIST SP 800-53** and **FedRAMP**, directly from vulnerability findings rather than from control failures. Built at an internal hackathon and **adopted by the Now on Now support team**, who ran it against ServiceNow's own instance.
- **Hackathon Projects**:
  - **Diagnostic Framework** — internal support tooling that streamlined issue tracking and reproduction across customer instances, **reducing debugging time by 40%**.
  - **ML over Vulnerability Response data** — Predictive Intelligence **clustering** to group vulnerable items beyond the product's filter-based group rules, and **regression** to predict time-to-remediate, where shipped remediation targets are deterministic date arithmetic.
  - **High-Volume Data Generator** — scalable generation of **100M+ records across 20–30 tables** for load testing; used by the performance team.

| Darwinbox Digital Solutions | Software Developer | Aug'18 – Mar'19 |
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- Designed and implemented a microservice architecture capturing an **audit trail** of client data modifications, written in **PHP with the Yii framework**, listening to **MongoDB oplogs** and publishing to an **AWS SQS** queue.
- Built an **Oplog Processor in Node.js** diffing strings, arrays and embedded documents into human-readable audit entries, with an internal dashboard for adding trackable fields.

| NowFloats | Software Developer | Jun'17 – Jul'18 |
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- Primary work was the **Boost merchant portal's customer-relationship surface** — customer enquiries, call logs and call tracking — built in **C#/ASP.NET, Bootstrap, JavaScript, jQuery and Knockout.js** and serving **50,000+ merchant websites**.
- Integrated the **Facebook Graph API** so merchants could publish products and updates to their Facebook profile, page and shop (Ads extension).
- Built the **SPIDER** module using **machine learning** to predict sales revenue per representative from historical performance, with up to **84% accuracy**.
- Built an **offers and discounted-purchase flow** on **Kitsune**, NowFloats' serverless application framework, wiring in **k-Pay** and **Assured Purchase** — the escrow-backed payment-protection system that held buyer funds until an order was confirmed fulfilled — and surfaced it to merchants through the **Boost Keyboard**; also built the **Order Management** dashboard (**AngularJS, C#**) covering purchase, payment and delivery status and grievance redressal.
- Worked on the **appointment-scheduling system** across the Kitsune-built merchant websites — per-staff availability and slot modelling, booking and management — in **Node.js and C#**, generic across the clinic, salon, restaurant and hotel verticals, where it ships today as an appointment scheduler for practices and a room-reservation engine for hotels.

| **Finomena** (wound down 2017) | Software Developer | Feb'17 – May'17 |
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- Built the internal dashboards for **Loan Management**, mandate (auto-debit) management, collections and transaction tracking on the small-ticket lending platform, in **PHP (Laravel)**, scaling to **8,000+ loan applications/month** with Eloquent ORM over MySQL.
- Used **Artisan** for task scheduling, cron handling and testing; Laravel modules for authorisation, validation, event handling and mailing.

| **Babajob.com** (acquired by Quikr, Jun'17) | Software Engineer | May'16 – Jan'17 |
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- Built **TelephonySystem**, the **Golang** service behind the missed-call job channel and consumed by Babajob's C# platform — inbound and outbound call handlers against **Exotel** numbers, call-rotation strategy, incoming-call handling and the logging subsystem. **Stateful by design**: a first call created the jobseeker's profile, and later calls were recognised by source number and enriched it — so a feature-phone user built a profile by ringing a number. **10,000+ inbound calls and 20,000+ outbound SMS per day** on **RabbitMQ**, **Redis** and **MongoDB**; one of two developers on a five-person telephony team.
- Wired that profile into an **automated matching and application pipeline**: each completed call published the jobseeker profile to the outbound messaging service, which matched it against open roles through **Elasticsearch**, sent the shortlist by SMS, then walked the seeker through a sequential application that connected them to the job creator.
- Contributed to the **search subsystem in Java over Elasticsearch**, with dynamic query generation via Mustache templates.

## Selected Projects — 2025 to present

Independent and consulting engineering through **Tathya Dev** (tathya.dev) — **10+ production web products across 43 repositories and 4,238 commits in 12 months**, enabled by AI-augmented engineering workflows.

| **AI & agent tooling** | |
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| **propagate**<br>Claude Code plugin · Node, ESM | • Sole-authored: **218 commits over six weeks**, 313 source files, **142 tests** — github.com/Rupali59/propagate. Declares the couplings between files that must change together and derives drift from file *content* on demand rather than storing state, currently watching **~920 edges across 14 workspaces**. Installed at user level, so its hooks execute in **every agent session on the machine** rather than only where it is checked out. A companion marketplace serves **18 skills across 3 plugins**.<br>• Built a `PreToolUse` hook that pattern-matches the literal command an agent is about to execute against declared hazards and surfaces the matching one *before* it runs; written after a documented hazard sitting three files away was hit anyway and cost **11 false verification records**. Added a companion `SessionStart` hook maintaining **18 canonical engineering rules**, and a restatement detector tracks them across **50 `CLAUDE.md` files** — it counts a file that restates a rule *even where the file also references it*, because a pointer beside a stale copy is what a half-finished migration looks like.<br>• Shipped a `selftest` with every rule detector and both pattern-matching guardrails that runs each trigger against a known-positive and asserts it fires, plus negative cases it must *not* match, because **a check that cannot fail reports success**; it caught a rule whose detector could never have matched its own body. A Claude Cowork job-classification skill is tested against captured fixtures.<br>• Designed it so no downstream file is ever edited automatically — the tooling finds and explains, a person approves, and each decision is recorded as a ledger event. The design principle throughout is that **"found nothing" and "looked at nothing" must be different outputs** — a probe that cannot report its own failure reports success. Auditing my own monitor against that standard found it logging a hardcoded success field while **49 crashes sat in a stderr log nothing read**; the logger could not express the state it was being cited to prove. |

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| **LLM Classification Pipeline**<br>Demand-intelligence ingestion · Gemini · schema-constrained | Built a production LLM classification pipeline, constraining Gemini calls with enforced response schemas at `temperature: 0` and batched, with determinism resting on cache-on-first-write rather than on the model, and a keyword-classifier fallback so a provider outage degrades quality instead of breaking ingestion. Model-proposed sources are **existence-checked against the live API before acceptance**, and an inconclusive check returns the candidate flagged rather than asserted. |
| **Products & platforms** | |
| **Multi-tenant CRM & Workflow Platform**<br>Shree Shyam Jyotish Kendra · ssjk-crm.vercel.app · Express · Next.js 14 | Designed and built an Express API and two Next.js 14 applications (client and admin), with a **database-driven RBAC/ACL engine** — runtime-editable roles, generic field-level access control over CRUD, per-actor tier projection, table→field cascade, and a capability-matrix UI whose review-diff modal shows the consequences of a change before an atomic batch save, with fail-closed reconciliation and admin-floor guards. WebAuthn + OTP authentication, BullMQ/Redis job queue, workflow engine, web-push. Phased task-model migration dark-shipped in five stages with a one-shot backfill of **~14,000 records**. Test infrastructure cut from **104s to 28s** across two changes — one shared `mongod` for the suite instead of 89, then per-suite index scoping, which is what actually moved the number. **228 test files.** |
| **Content & SEO Platform**<br>Vipin Kaushik · vipinkaushik.com · Next.js · TypeScript | Shipped a production Next.js site backed by the computation service below — the two are one system. Nine templated landing pages plus a hub on a shared shell with **FAQPage structured data** and selective `noindex`, a server-side analytics pipeline, signed-cookie location handling with a durable resolved-location cache, and a migration of public pages to **static generation**. Largest single build of the last quarter. |
| **Demand-Intelligence Platform**<br>Multi-source ingestion & scoring · Next.js · Mongoose · Gemini | Built multi-source ingestion and opportunity scoring over Trends, YouTube and Search Console, fed by the schema-constrained Gemini classification pipeline described above, with an entity ontology mapping demand against supply. Search Console ingestion authenticated via **Workload Identity Federation**, adopted to remove service-account key fragility; scheduling moved off hosted cron onto a local agent. |
| **Ephemeris Computation Service**<br>Swiss Ephemeris over FastAPI · Python 3.12 · MongoDB | Engineered a FastAPI service over Swiss Ephemeris, a native library with process-global mutable state that is **not thread-safe**, serialising every call through a single-worker executor that releases the event loop during computation. Adds an astronomical-timeline subsystem (boundary and instant finders, lunar-month tables, per-year caching with TTL) and typed per-calculation endpoints replacing one overloaded route. **1,034 tests.** |
| **Motherboard**<br>API gateway & plugin platform · Go 1.25 · Next.js · Docker | Architected a Go multi-module monorepo implementing an API gateway and multi-tenant plugin platform — **25 `go.work` modules, 652 tracked Go files**. Plugin services authenticate, publish their routes to a client manager, and the gateway proxies to them behind a **fail-closed entitlement gate** and a plugin-key claim assertion. |
| **E-commerce Storefront & Embedded Admin**<br>Gemastrology · Shopify · Liquid · Remix | Developed a Shopify Liquid theme and an embedded Remix admin application — metafield/metaobject schema installer, headless catalogue seeders, and certificate verification implemented natively in Liquid rather than through an app proxy. Accessibility pass to a **44×44 touch-target floor**. |
| **Production Portfolio Sites**<br>Manav Daehi · Kirti Kumar · Youvan Prakashan · Next.js · React 19 | Shipped three production sites. manavdaehi.com — built a Google Docs→works sync pipeline and multi-codec delivery (AV1/HEVC/H.264); a hero codec ladder took **10.4 MB to ~3 MB**, with delivery suppressed for non-watchers and an H.264 Level 4.0 re-encode for iOS autoplay. kirtik.co.uk — dual-track portfolio with Playwright end-to-end coverage. rtistic.vercel.app — Next.js 14 and MongoDB, jest suite and CI, public source. youvanprakashan.org — publishing platform with an Express admin classification queue and a language classifier. |
| **Git Forensics & Recovery**<br>Incident work · git | Recovered **154 commits** left unreachable in a live repository after a worktree removal combined with dropped stashes — located, recovered and permanently tagged all 154 under `refs/tags/rescue/*`; **8 of 29 stashes** held content present in no branch, and the restored work was reintegrated behind its tests. Separately diagnosed a repository whose `origin` pointed at an **unrelated project with no common ancestor** — 48 commits and 341 files that a single `git push` would have sent to the wrong remote. |
| **Text Digitisation & Translation Pipeline**<br>Classical-text corpus · Sanskrit-texts · Python | Built ingestion for a structured-JSON corpus of classical texts: a TEI/SARIT parser handling prose and division-type detection, a scraper splitting on the source's own headings, a verse-marker grammar, and per-work structural validators. One structural fix took depth errors in a single work from **3,522 to 6**. Corpus spans **90,000+ verses across 50+ texts**. |

## Dissertation & Coursework

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| **Dissertation** | Human Activity Recognition using an **HMM-based Intermediate Matching Kernel**, representing videos as sequences of sets of feature vectors. Re-implemented in 2026 as a containerised Python pipeline with a Flask front end — github.com/Rupali59/himk-implementation. |
| **Methodology** | Classifies a video into a predefined activity class. Feature extraction and classification in Python with scikit-learn, hmmlearn and OpenCV; originally prototyped in MATLAB with Piotr's CV Toolbox, PNetCDF and OpenMPI. |
| **ML Techniques** | Bayesian classifiers, Gaussian mixture models, perceptrons, artificial neural networks and support vector machines on linearly separable, non-linearly separable and overlapping data. |

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| **Presentations** | Presented the components of an information retrieval system and Google's PageRank model at the departmental seminar on Information Retrieval. |
| **Other Projects** | Management Information System (MIS), Cybernetics Protector System, Product Management System. |

## Technical Skills

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| **Languages** | Go, TypeScript, JavaScript, Python, PHP, Java, C#, C++, GlidePlatform |
| **Web & Frameworks** | Next.js, React, Node.js, Express, Remix, Astro, Laravel, .NET, FastAPI, Tailwind, Bootstrap |
| **AI & Agent Tooling** | Claude Code plugins, hooks and skills; structured LLM output and schema-constrained generation; eval harnesses and fixture-based regression suites; hallucination containment and graceful degradation; human-in-the-loop design; MCP |
| **Databases** | MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch |
| **Queues & Messaging** | RabbitMQ, BullMQ, AWS SQS, Redis Queue |
| **Cloud & DevOps** | AWS, Vercel, Fly.io, Docker, Shopify, Jenkins, Git, Linux |
| **ML** | scikit-learn, hmmlearn, OpenCV, MATLAB/Octave, pattern recognition and classification |

## Academic Profile

| Year | Degree | Discipline | Institution | Score |
|---|---|---|---|---|
| 2016 | **M.Tech** | Computer Science & Engineering<br>Pattern Recognition & Machine Learning | National Institute of Technology, Goa | 8.42 CGPA |
| 2014 | **B.E.** | Computer Science & Engineering<br>Graduated with Honours | LNCT, Bhopal (RGPV) | 8.58 CGPA |
| 2010 | XII (CBSE) | Science | St. Joseph's Convent School, Bhopal | 83.0% |
| 2008 | X (CBSE) | — | St. Joseph's Convent School, Bhopal | 88.0% |

## Certifications

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| **AI — 2026** | **Anthropic:** Claude Code in Action; Introduction to Claude Cowork; Claude Platform 101; Claude Code 101; Claude 101. **Google:** Gemini Enterprise Agent Ready; Create Your First Gemini Enterprise Application; Google Cloud Innovator; Developer Program Premium Tier; AI Fundamentals (Google Career Certificates). **Verify:** credly.com/users/rupali-bhatnagar-59 · g.dev/rupali59. |
| **Earlier** | WhatsApp API Expert, Wati (2024) · Ultimate JavaScript: Fundamentals, Code With Mosh (2025) · Neural Networks & Deep Learning, Coursera (2020) · JavaScript Algorithms & Data Structures, freeCodeCamp (2020) · Git Essential Training, LinkedIn Learning (2020) |

## Achievements & Activities

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| **Professional** | • Winner, NowFloats Hackathon for product innovation in sales — built the ML-powered SPIDER revenue-prediction module. 2018<br>• **Geektrust GOLD** — score **95.8** on the Metro Card backend challenge, with **Readability** and **Maintainability** assessed at **ADVANCED** tier by an independent reviewer. |
| **Academic** | • **Chancellor's Scholarship (Topper)**, Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — awarded to the top 10 students statewide by theory marks. 2012, 2013<br>• **Silver Medal**, Persistent Systems Limited Programming Contest, inter-college. 2013 |
| **Other** | • Organised the intra-college programming competition, LNCT Bhopal. 2012<br>• Conducted cultural events, NIT Goa. 2014–2016 |

## Positions of Responsibility

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| **NIT Goa** | Teaching Assistant, Computer Science & Engineering Department | 2014–2016 |
| **NIT Goa** | Vice President, Computer Science Department | 2014–2016 |
| **LNCT, Bhopal** | Student Coordinator, Training & Placement Department | 2013–2014 |